AP CALCULUS AB

Unit 1 • Lesson 1.2 practice • Answer key

# Lesson 1.2 • Defining Limits and Using Limit Notation

4 multiple-choice questions and 1 free-response question • Answer key with explanations

### Question 1

What does $\lim_{x \to 2} f(x) = 5$ mean?

(A) $f(2) = 5$.

(B) $f(x)$ gets as close to 5 as we want when $x$ is close enough to 2, $x \neq 2$.

(C) $f(x)$ never equals 5.

(D) $f(5) = 2$.

**Solution**

**Answer: (B).** A limit describes outputs near $x = 2$, not the value at $x = 2$.

### Question 2

$\lim_{x \to 3^-} f(x) = 4$, $\lim_{x \to 3^+} f(x) = 4$ and $f(3) = 1$. What is $\lim_{x \to 3} f(x)$?

(A) 1

(B) 4

(C) 5

(D) It does not exist.

**Solution**

**Answer: (B).** Both one-sided limits equal 4. The value $f(3)$ does not matter.

### Question 3

$\lim_{x \to 0^-} g(x) = 2$ and $\lim_{x \to 0^+} g(x) = 5$. Which statement is true?

(A) $\lim_{x \to 0} g(x) = 2$

(B) $\lim_{x \to 0} g(x) = 3.5$

(C) $\lim_{x \to 0} g(x)$ does not exist.

(D) $g(0) = 5$

**Solution**

**Answer: (C).** The one-sided limits are different.

### Question 4

Let $f(x) = x^2$ for $x \neq 1$ and $f(1) = 7$. What is $\lim_{x \to 1} f(x)$?

(A) 7

(B) 8

(C) It does not exist.

(D) 1

**Solution**

**Answer: (D).** Near $x = 1$ (but not at it), $f(x) = x^2$, which approaches 1.

### Question 5 (Free response)

Let $f(x) = \begin{cases} x + 2, & x < 1 \\ 5, & x = 1 \\ x^2 + 2, & x > 1. \end{cases}$

(a) Find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$.
(b) Does $\lim_{x \to 1} f(x)$ exist? If so, give its value.
(c) Explain why the value $f(1) = 5$ does not change your answer to (b).

**Solution**
(a) $\lim_{x \to 1^-} f(x) = 1 + 2 = 3$ and $\lim_{x \to 1^+} f(x) = 1^2 + 2 = 3$.
(b) Yes. Both one-sided limits equal 3, so $\lim_{x \to 1} f(x) = 3$.
(c) A limit only uses values of $f(x)$ for $x$ near 1 with $x \neq 1$; the value at $x = 1$ itself is not used.